



Agnel Charities'

Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

Computer Engineering Department

Major Project Progress Presentation-II

B.Tech in Computer Engineering Sem - VII

2026-27

Intelligent Diabetes Clinical Decision Support System



Group No. 28

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Presentation Outline



- Problem Statement
- Proposed System and scope
- Hardware & Software Requirements
- Timeline and Gantt chart (interaction sheet)
- System design and working of the system
- Datasets (synthetic and real Indian data)
- Implementation and screens
- Initial Results / Demo
- Work Completed vs Planned, progress
- Conclusion



Problem Statement

- After **metformin**, doctors choose between **SGLT2 inhibitors**, **DPP-4 inhibitors** and **sulfonylureas**
- **Question:** which option lowers HbA1c most for **this patient**?
- **Old records mislead:** different patients get different drugs

Starting HbA1c	Got SGLT2 inhibitor	Got sulfonylurea	Difference
High	300 patients, -1.2	100 patients, -1.0	-0.2
Lower	100 patients, -0.6	300 patients, -0.5	-0.1
All patients (naive)	-1.05	-0.625	-0.425

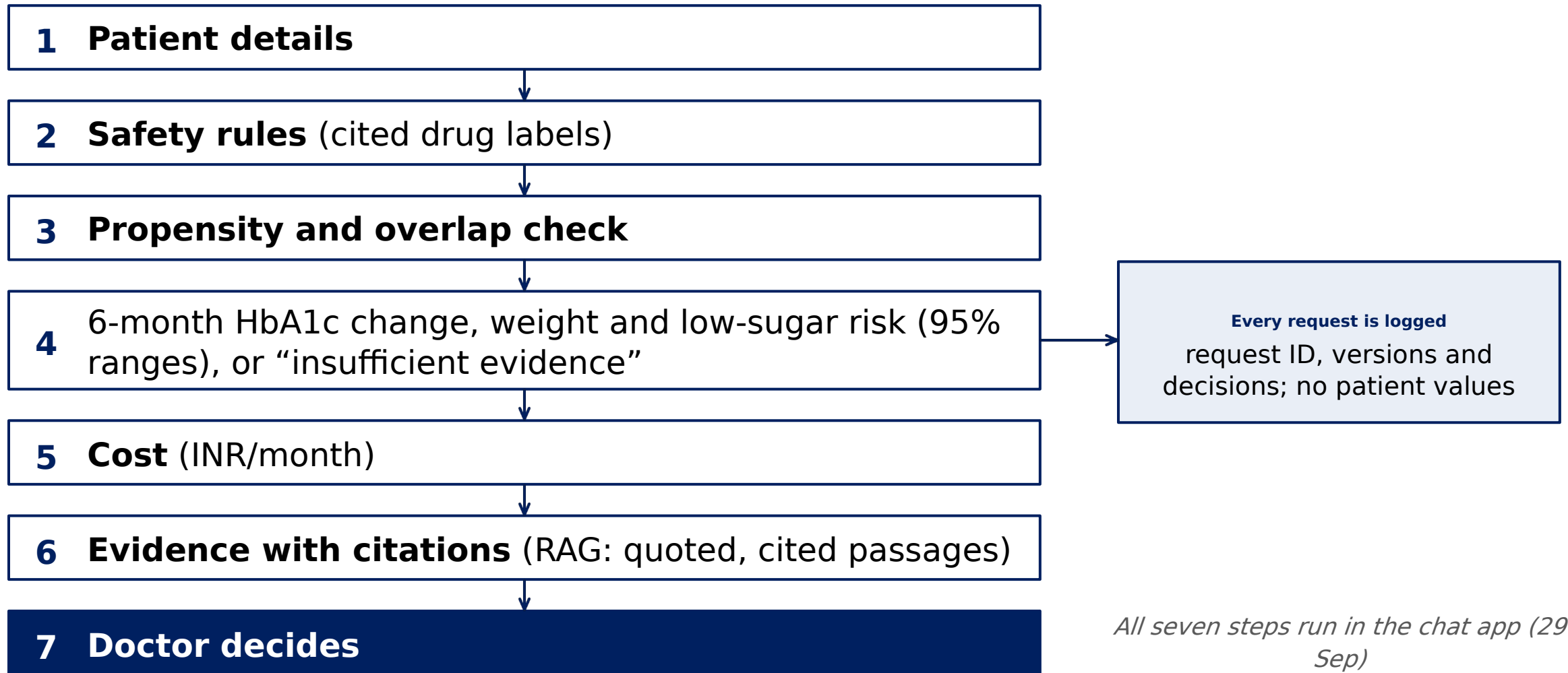
Naive difference -0.425 vs fair difference -0.15 (about 3× too big)

Fair = compare like with like, then average: $\frac{1}{2} \times (-0.2) + \frac{1}{2} \times (-0.1)$

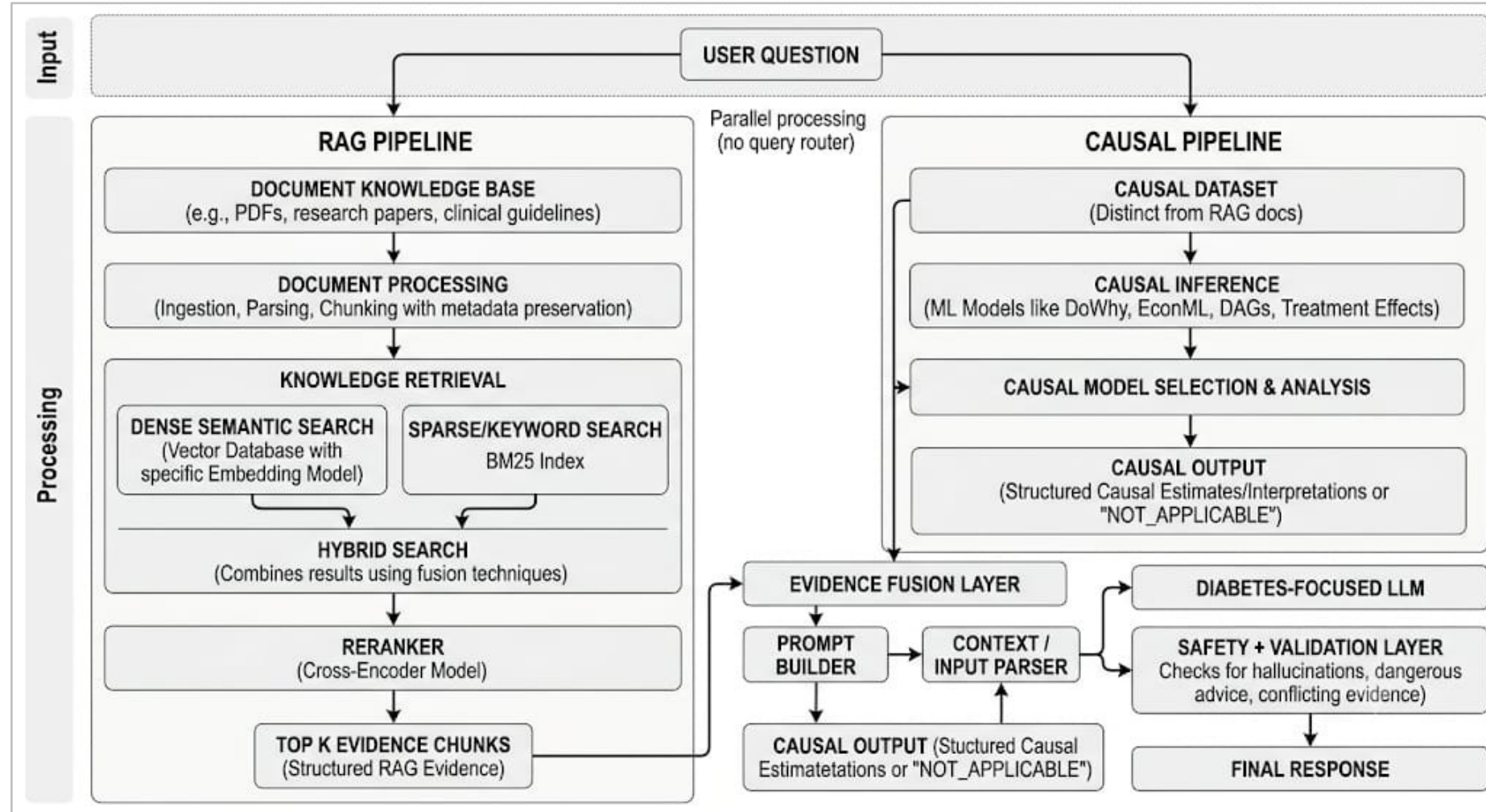
Illustrative synthetic numbers; HbA1c change in %



Proposed System – How DiaCausal Works

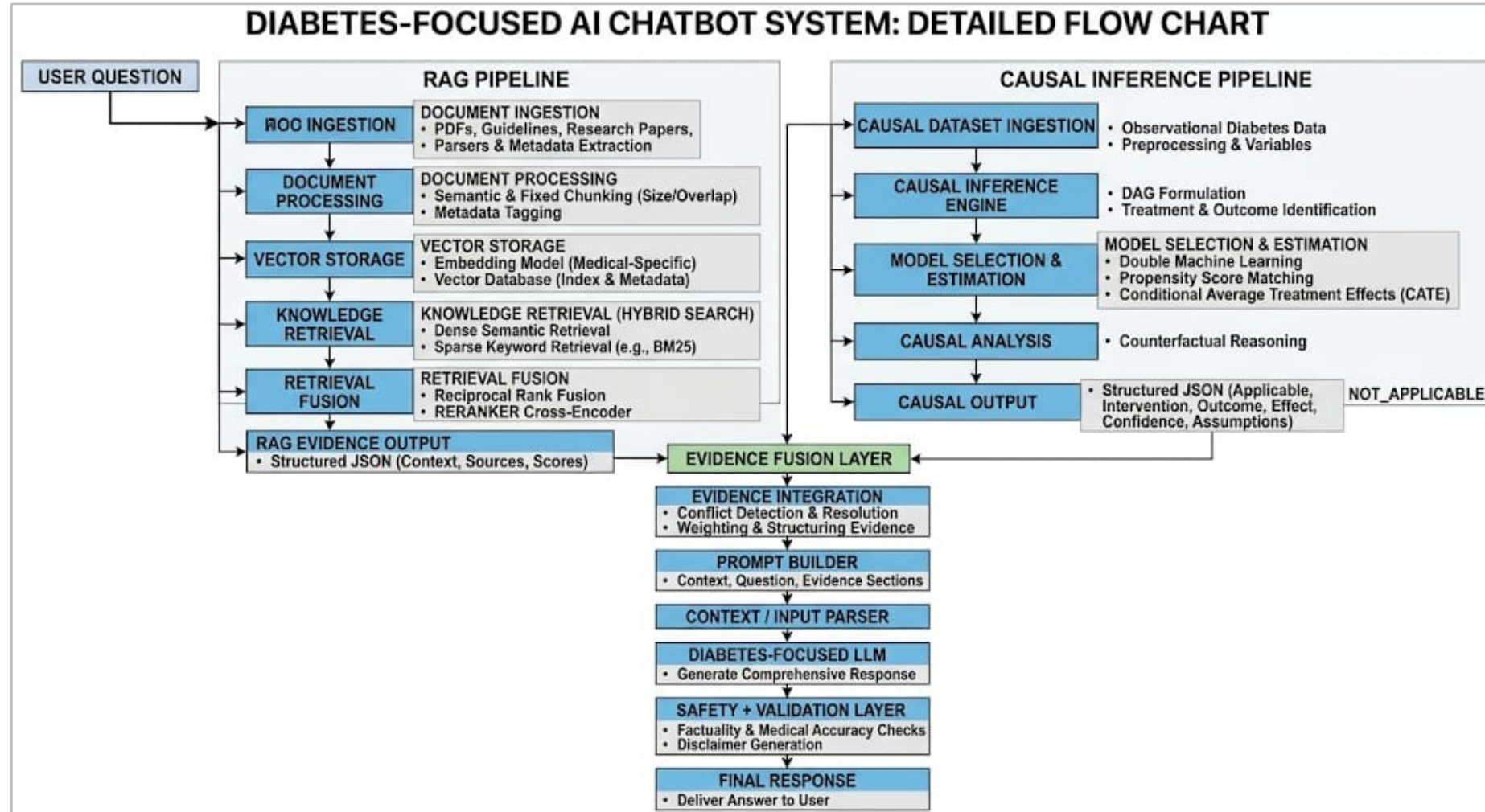


Proposed System – Block Diagram



RAG and causal pipelines run in parallel, then meet in evidence fusion

Proposed System – Flowchart



Every stage from ingestion to final response, with safety checks



Proposed System – Safety Rules

- 1 The **doctor always decides**
- 2 **Safety rules run before** any estimate
- 3 **Doses and thresholds** only from cited tables, never from AI
- 4 **Refuse to guess** (“insufficient evidence”)
- 5 Every estimate has a **95% range**
- 6 Every request is **logged**

Intended use: Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.



Proposed System – Scope

In scope (built and tested)	Out of scope (the system says so and stops)
Adults with type 2 diabetes already on metformin	Type 1 diabetes, pregnancy, under-18s
Three add-on classes: SGLT2i, DPP-4i, sulfonylurea	Starting insulin; emergencies (DKA, HHS)
6-month HbA1c change, weight, low-sugar risk, each with a 95% range	Drug doses (only the label and the doctor)
Safety rules first: kidney function, pancreatitis, hypoglycaemia	Diagnosis or prescribing: the doctor decides
Cited evidence from licence-cleared sources (WHO, FDA)	Patient identifiers: Aadhaar, phone, PAN, email removed
Clinician users: website, chat app, REST API	Real patient data until ethics approval

Future scope: GLP-1 receptor agonists and pioglitazone · Indian prices (Jan Aushadhi) · hospital records with ethics approval · Hindi and Marathi interface



Hardware & Software Requirements

Tier	Hardware	Software
Development	Laptop with 8 GB+ RAM (MacBook Air, Apple silicon)	Python 3.12, NumPy, pandas, scikit-learn, Matplotlib, pytest, Git/GitHub, Claude Code
Demo	Same laptop, offline; or the website (diacausal.netlify.app)	Streamlit, FastAPI, Uvicorn; website on Netlify with Supabase accounts
Client	Laptop or phone	Any modern web browser
RAG (built)	Same laptop	PDF text, BM25 + TF-IDF search, quoted cited answers; optional Gemini or Ollama

Minimum: 8 GB RAM and 5 GB free disk · the demo runs fully offline



System Requirements

Functional

Requirement	Status
Patient input with unit checks	Built
Safety rules with cited reasons	Built
Estimate with 95% range, or “insufficient evidence”	Built
Hypoglycaemia risk and weight change	Built
Cost lookup (INR/month)	Built*
Audit log	Built
REST API	Built
Evidence with citations (RAG)	Built
Website with accounts (MFA)	Built

**Prices show “unavailable” until confirmed*

Non-functional

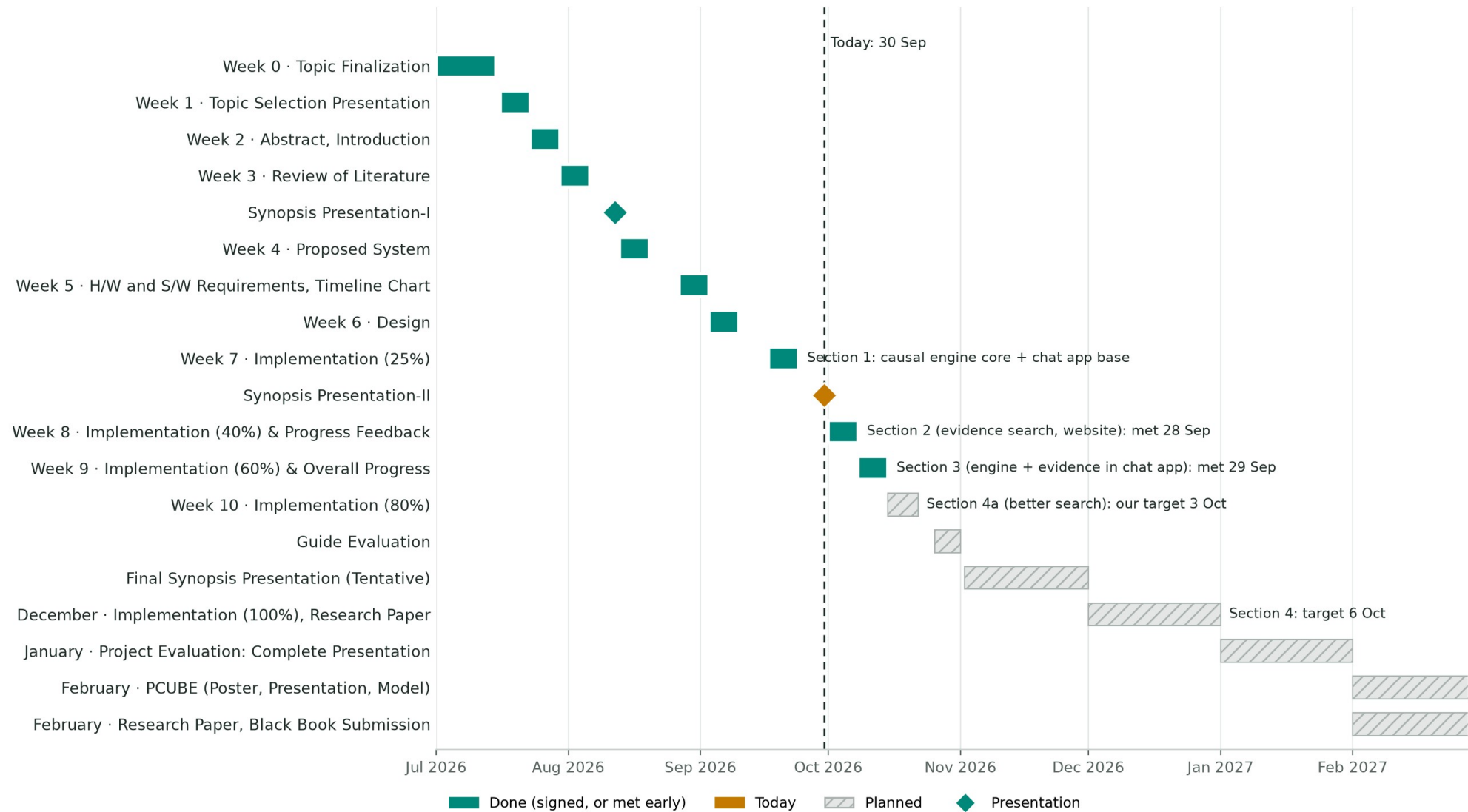
Quality	Target
Speed	Under 3 s per request
Logging	100% of requests logged
Testing	At least 80% test coverage
Reproducible	Fixed random seeds
Intended use	Statement on every screen
Accessible	WCAG 2.1 AA

Measured: about 12 ms per request. Coverage and WCAG not yet measured.



Timeline / Gantt Chart

Gantt chart: Project Interaction Sheet, Group 28 (July 2026 - February 2027)





Timeline Chart – Interaction Sheet

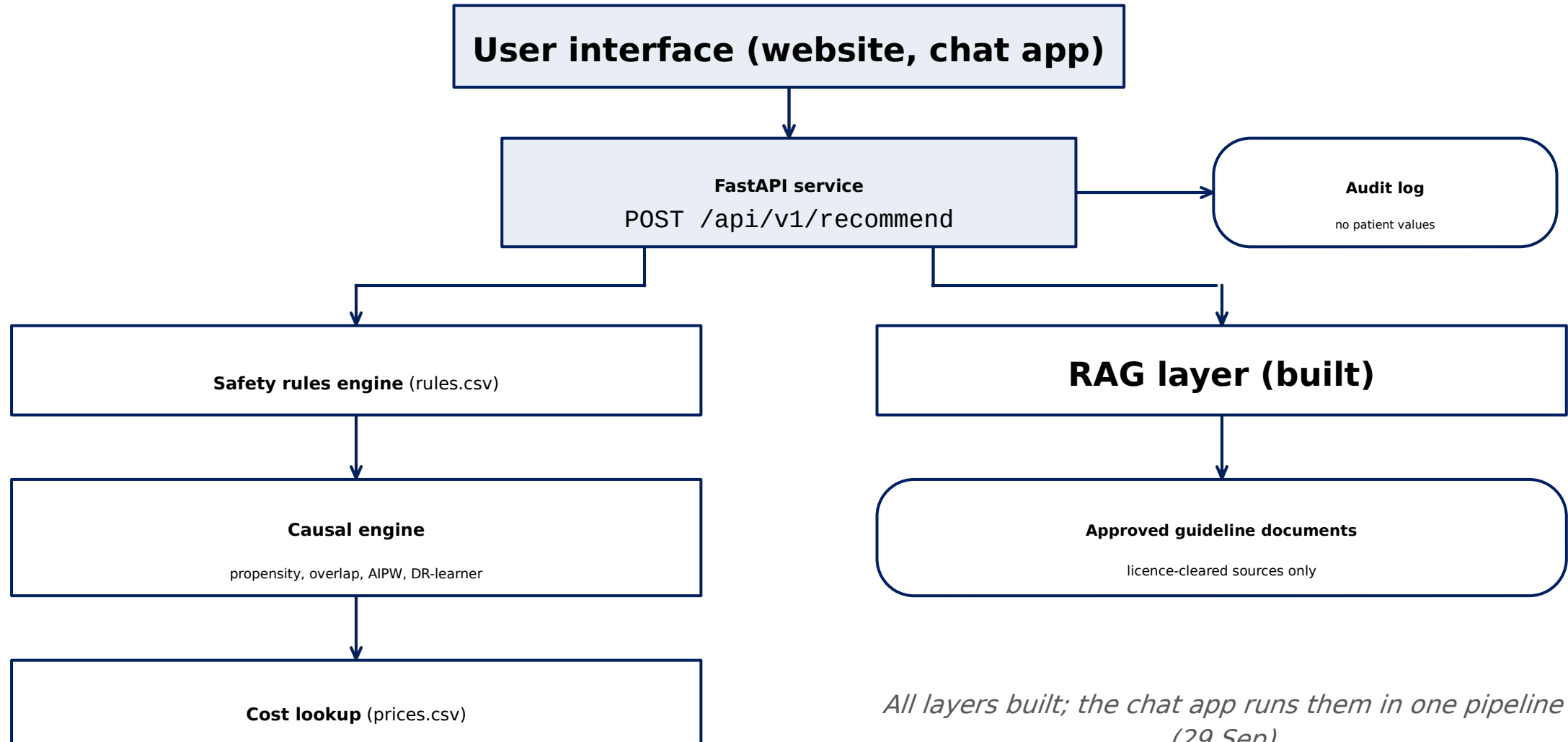
Timeline chart: Project Interaction Sheet, Group 28 (2026–27)

Week 0		● Topic Finalization	Signed
Week 1	22/07/2026	◆ Topic Selection Presentation	Signed
Week 2	29/07/2026	● Abstract, Introduction	Signed
Week 3	05/08/2026	● Review of Literature	Signed
	12/08/2026	◆ Synopsis Presentation-I	Done
Week 4	19/08/2026	● Proposed System (define problem, approach, scope)	Signed
Week 5	02/09/2026	● H/W and S/W Requirements & Timeline Chart	Signed
Week 6	09/09/2026	● Design	Signed
Week 7	23/09/2026	● Implementation (25%): Section 1, causal engine core + chat app base	Signed
	30/09/2026	◆ Synopsis Presentation-II	Today
Week 8	07/10/2026	● Implementation (40%) & Overall Progress Feedback: Section 2	Met 28 Sep
Week 9	14/10/2026	● Implementation (60%) & Overall Progress: Section 3	Met 29 Sep
Week 10	21/10/2026	○ Implementation (80%): Section 4a, better search	Target 3 Oct
	26-31/10/2026	◇ Guide Evaluation	Planned
	November 2026	◇ Final Synopsis Presentation (Tentative)	Planned
	December 2026	○ Implementation (100%): Section 4 complete; Research Paper	Target 6 Oct
	January 2027	◇ Project Evaluation: Complete Project Presentation	Planned
	February 2027	◇ Project Evaluation: PCUBE (Poster, Presentation and Model)	Planned
	February 2027	○ Research Paper Revision, Submission; Black Book Submission	Planned

Implementation in four sections (sheet note): 1 causal engine core · 2 evidence search (RAG), secondary outcomes, website · 3 causal engine + evidence inside the chat app · 4 better search, real-data adapter, evaluation tools.



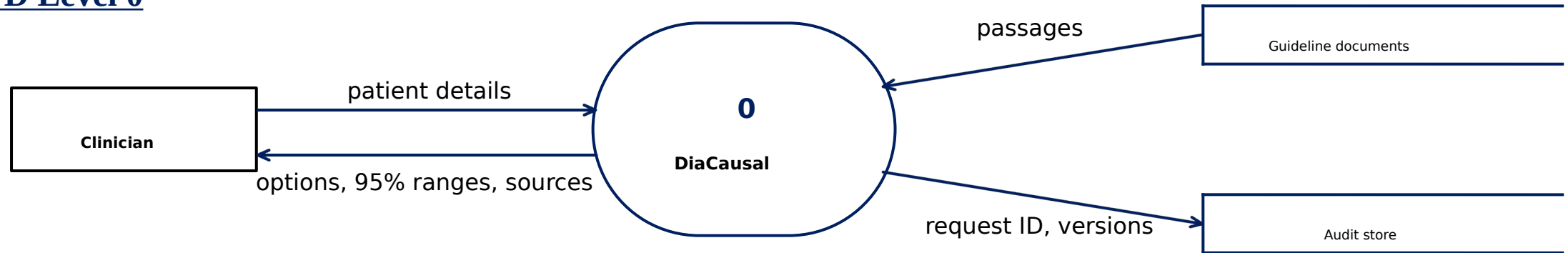
System Design – Architecture



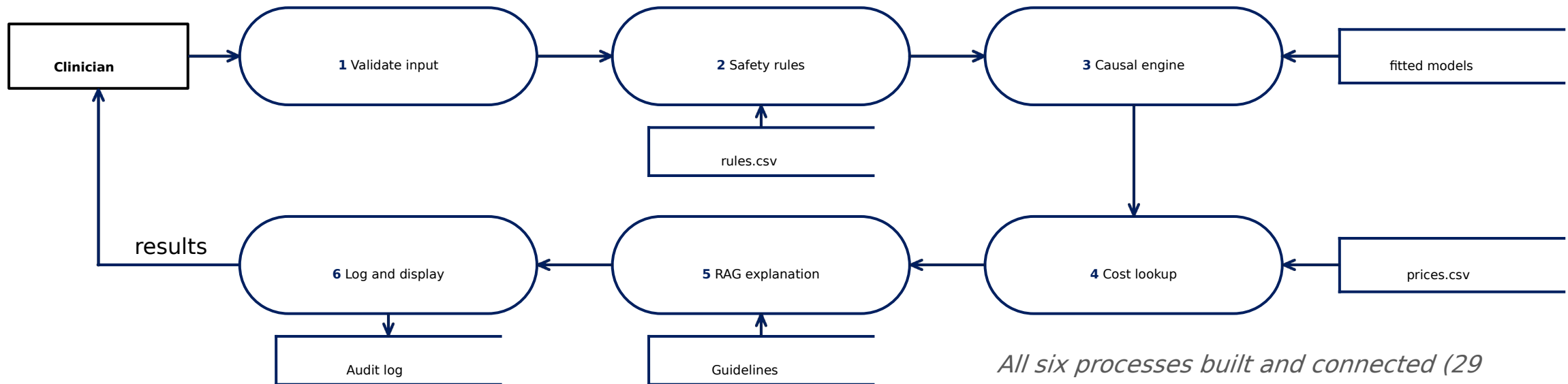
*All layers built; the chat app runs them in one pipeline
(29 Sep)*

System Design – Data Flow

DFD Level 0

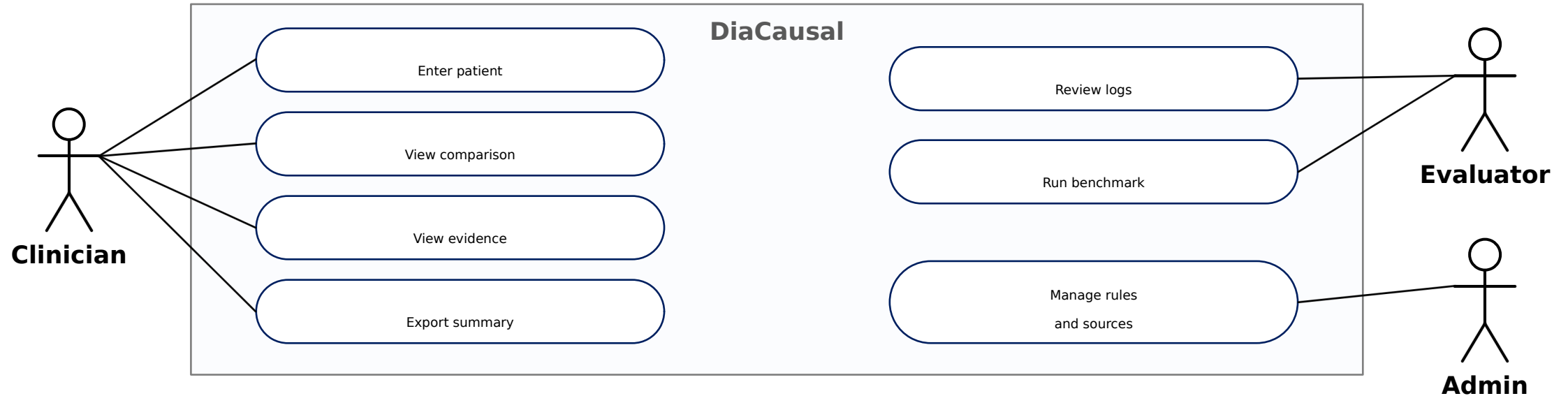


DFD Level 1

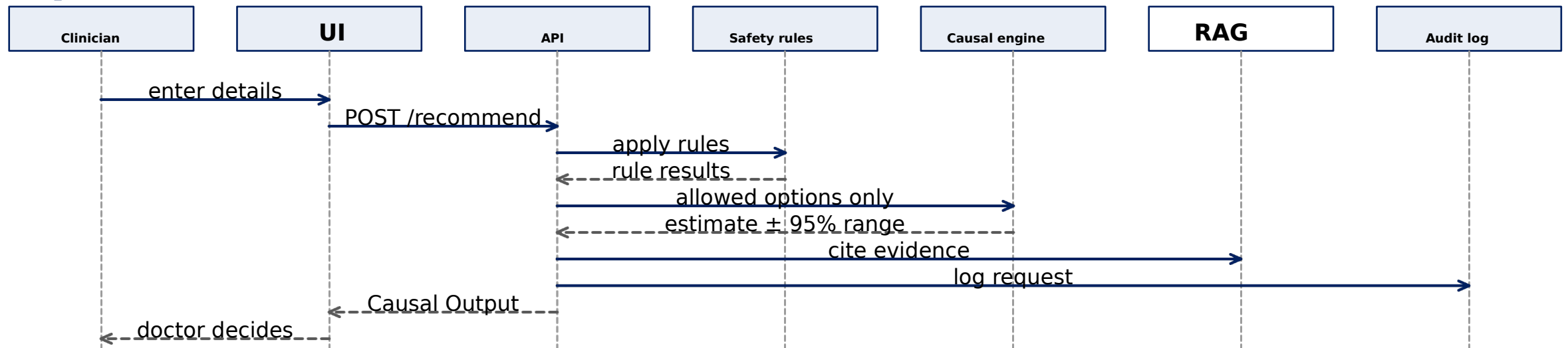


All six processes built and connected (29 Sep)

System Design – Use Case and Sequence



Sequence: one consultation





System Design – Working of the System

Step	What happens	For this patient (demo preset 2)
1 Input	Check units and plausible ranges; Asian-Indian BMI category	60-year-old woman, HbA1c 8.2%, eGFR 40, BMI 25.5 (obese $\geq$ 25), past pancreatitis
2 Safety rules	Cited rules run before any estimate	SGLT2i excluded (R01: eGFR < 45); DPP-4i caution (R04, R05); SU caution (R09)
3 Overlap	Enough similar patients got each drug?	DPP-4i 0.54, SU 0.23: both above 0.05, so estimates are allowed
4 Estimate	DR-learner: 6-month HbA1c change, 95% range	DPP-4i -0.68 (-1.09 to -0.27); SU -0.82 (-1.21 to -0.42); ranges overlap
5 Also	Weight, low-sugar risk; evidence; cost	Weight $+0.6$ kg for both; cited FDA passages; price unavailable
6 Decide, log	Doctor decides; audit keeps IDs and versions only	Request ID, engine 0.3.0, rule IDs R01 R04 R05 R09; no patient values

Numbers from the causal engine for this patient (synthetic data); the same flow runs in the website, the API and, since 29 September, the chat app.



System Design – Data and API

rules.csv – safety rules (R01–R10, each cited)

rule id	drug class	condition	action	message	source	section	status
R01	SGLT2i	eGFR < 45	EXCLUDE	Not recommended to improve glycaemic control when eGFR < 45	FARXIGA label; KDIGO 2022	Sec 1, 2.2	VERIFIED
R05	DPP-4i	past pancreatitis	CAUTION	Not studied in patients with a history of pancreatitis	JANUVIA label	Sec 1; 5.1	VERIFIED
R08	SU	age ≥ 65	CAUTION	Higher hypoglycaemia risk; age cut-off 65 is team-set	Glimepiride label	Sec 2.1; 8.5	UNVERIFIED

prices.csv

drug	INR per month	as-of date	source
dapagliflozin (SGLT2i)	price unavailable	—	to be confirmed (Jan Aushadhi list)

POST /api/v1/recommend

Input: patient details

```
{"age": 60, "sex": "female",  
  "duration_years": 8, "hba1c": 8.2,  
  "egfr": 40, "bmi": 25.5,  
  "pancreatitis_history": true}
```

Output: Causal Output

- Each option: estimate with 95% range, or “insufficient evidence”
- Exclusions and cautions, each with its source
- Versions and a request ID



Datasets – What We Use

Dataset	Real or synthetic	Size	Used for
Synthetic patient cohort	Synthetic, India-calibrated	5,000 patients × 20 cohorts	Train and grade the causal engine (true answer known)
Safety rules (rules.csv)	Real: drug labels, KDIGO	10 rules	Remove unsafe options before any estimate
Generator parameters (params.yaml)	Each number: source + status	117 numbers	How the synthetic patients are made
Evidence sources (RAG)	Real: WHO 2018, 7 US FDA notices	26 listed, 8 in the search	Cited evidence and quoted answers
Evidence test questions	Written by the team	60 questions	Measure the evidence search
NMB-2017 survey (new)	Real Indian data, CC0	7,496 adults (1,986 diabetic)	Check that synthetic patients look real

No public Indian dataset records which add-on drug each patient got and their HbA1c 6 months later, so the engine is trained and graded on synthetic data, where the true effect is known.



Dataset – Synthetic Cohort (How It Is Made)

- 1 Draw 5,000 adults on metformin with HbA1c $\geq 7\%$: age, sex, years with diabetes, HbA1c, eGFR, BMI, history
- 2 Give each a drug the way doctors tend to (sicker or heavier patients get different drugs = confounding)
- 3 Work out each patient's 6-month HbA1c change under all 3 drugs (directions from published studies)
- 4 Keep only the drug they got; the other two stay hidden, and that hidden truth lets us grade every method
- 5 Weight change and low-sugar risk made the same way, from a separate random stream

1. Synthetic cohort: first 8 of 5,000 patients (SYNTHETIC, observed columns only)

age	female	duration_years	hba1c	egfr	bmi	ascvd	hf	hypo_history	low_income	ckd	treatment	y	weight_change_6m	hypo_6m
56	1	8.60	7.70	52	22.40	0	0	0	0	1	SU	-1.38	3.00	0
54	1	11.90	9.70	85	28.30	0	0	0	1	0	SGLT2i	-1.55	-3.48	0
61	1	0.80	9.80	76	27.80	0	0	0	0	0	SU	-1.07	2.18	0
56	1	4.20	9.70	85	22.60	0	0	0	0	0	SGLT2i	-0.37	-0.24	0
50	0	7.00	9.70	112	30.40	1	0	0	0	0	SGLT2i	-1.76	-4.66	0
59	1	2.70	8.90	102	27.90	0	0	0	0	0	DPP4i	-0.90	4.18	0
68	1	3.00	9.10	78	23.70	0	0	0	0	0	SGLT2i	-1.12	-0.75	0
64	0	8.50	9.30	66	28.90	0	0	0	0	0	SU	-1.87	0.47	1

Generated by `diacausal_engine/cohort.py` from `data/params.yaml` (seed 0); columns `dka_history`, `pancreatitis_history` and `t1d` (always 0) not shown. One row = one patient on metformin who was given one add-on drug ('`treatment`'). `y` = 6-month HbA1c change (percentage points); `weight_change_6m` in kg; `hypo_6m` = any low-sugar episode (1 = yes). These are the only columns the estimators see; a real hospital record would have the same ones.



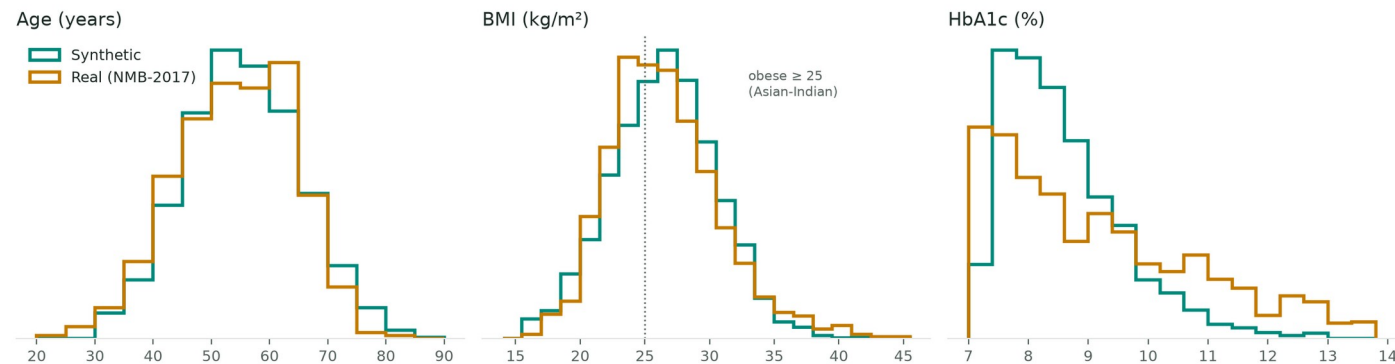
Dataset – Real Indian Data (NMB-2017) vs Synthetic

7. Real Indian data: NMB-2017, first 8 of 7,496 adults (REAL, CC0, reference only)

patient_id	state	district	age	gender	height_cm	weight_kg	waist_cm	diabetes_self_declared	diabetes_father	diabetes_mother	hba1c
0	Rajasthan	Ajmer	35	Female	157	47	70	Normal	0	0	4.90
1	Orissa	Balasore	38	Female	145	65	81	Normal	1	0	5.30
4	Rajasthan	Ajmer	42	Female	159	68	102	Normal	1	1	5.00
7	Orissa	Balasore	37	Female	142	44	76	Normal	1	0	5.40
8	Orissa	Balasore	47	Male	160	60	87	Normal	1	0	12.60
10	Karnataka	Bangalore	59	Female	158	57	83	Normal	0	0	6.30
13	Orissa	Balasore	30	Female	151	84	72	Normal	1	0	6.20
14	Orissa	Balasore	37	Female	151	60	81	Normal	1	0	4.90

Nagarathna et al. 2020, Diabetes Res Clin Pract; Mendeley Data doi:10.17632/twp8xw6p25.1 (CC0 1.0). Nationwide survey of adults at high risk of type 2 diabetes. No drug and no follow-up HbA1c, so it cannot train the engine; used to check the synthetic cohort looks real. Activity columns and zone not shown.

Synthetic cohort vs real Indian adults with diabetes and HbA1c $\geq 7\%$ (n = 1010)



Real: NMB-2017, Nagarathna et al. 2020, Mendeley Data doi:10.17632/twp8xw6p25.1 (CC0). It has no drug or follow-up data, so it checks realism only.

Matches real Indian adults:

age 55 vs 54, BMI 26.4 vs 26.6, women 45% vs 41%

Differs: HbA1c 8.6 vs 9.2

ours is narrower → widen it next

No drug or follow-up HbA1c:

realism check only, never used to train



Implementation – What's Built

Module	Status	Evidence
Synthetic patient generator	Done	cohort.py; every number sourced in params.yaml
Safety rules	Done	rules.csv: R01–R10, each with a source
Propensity and overlap check	Done	Abstains when propensity is below 0.05
Naive / IPW / AIPW	Done	estimators.py (plus matching); results table
DR-learner	Done	Per-patient estimate with a 95% interval
Metrics; weight and low-sugar risk	Done	Bias, RMSE, coverage, regret; weight and low-sugar risk
Benchmark	Done	20 cohorts × 5,000 patients; refutation 9 of 9
Demo app and website	Done	Streamlit; diacausal.netlify.app with accounts and MFA
API	Done	POST /api/v1/recommend
Automated tests	Done	150 engine, 33 RAG, 19 website; chat app 401 + 246
RAG pipeline	Done	WHO 2018 + 7 FDA notices; hybrid search; cited quoted answers



Implementation – Exactly What the 25% Covers

Step	What it does	Where in the code
1 Chat app front door	Sign-in (password, CAPTCHA, authenticator code), guards, patient panel, guardrails	backend/app/
2 Synthetic cohort	5,000 India-calibrated patients whose true drug effects are known	cohort.py, params.yaml
3 Safety rules first	Exclude or caution a drug from cited label rules before any estimate	rules.csv, guardrails.py
4 Propensity + overlap	Chance each patient got each drug; below 0.05 → “insufficient evidence”	propensity.py
5 Average effects	Naive vs IPW, matching, AIPW: removes the confounding bias	estimators.py
6 This patient's estimate	DR-learner: 6-month HbA1c change with a 95% range (HC3)	fitting.py, recommend.py

The 25% ends here: one patient in → unsafe drugs removed → each remaining drug’s 6-month HbA1c change with a 95% range, or “insufficient evidence” (Streamlit demo and API). Signed by our guide on 23 September.

Beyond the 25%: benchmark and 9 refutation checks · weight and low-sugar risk · evidence search (RAG) · website with accounts · engine + evidence inside the chat app (all six stages).



Implementation – How the Code Works

AIPW score (doubly robust) – estimators.py

```
def aipw_scores(Y, T, e, mu):  
    #  $\phi_a(X) = \mu_a(X) + 1\{T=a\} (Y - \mu_a(X)) / e_a(X)$   
    got = (T[:, None] == np.arange(len(ARMS))[None, :]).astype(float)  
    return mu + got * (Y[:, None] - mu) / e
```

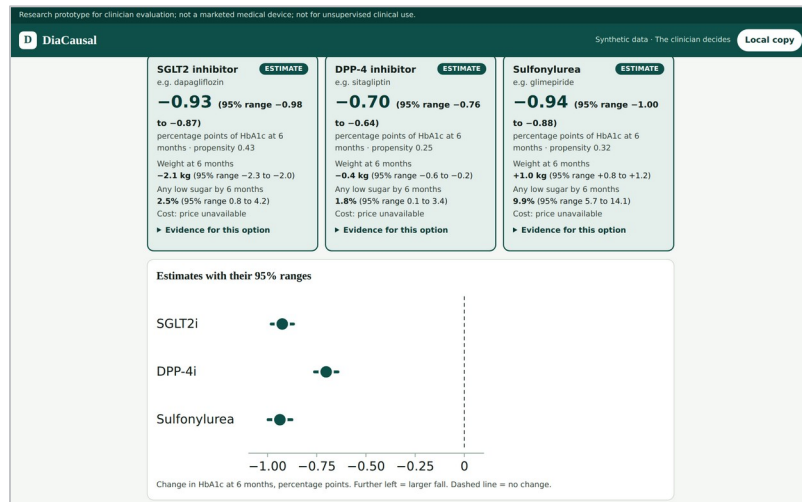
DR-learner: this patient's estimate with a 95% interval (condensed)

```
bread = np.linalg.pinv(B.T @ B)  
self.beta_ = bread @ B.T @ pseudo           # regress AIPW scores on details  
value = B @ self.beta_[ :, k]               # this patient's estimate  
se = np.sqrt(np.einsum("ij,jk,ik->i", B, self.cov_[k], B)) # HC3 SE  
low, high = value - z * se, value + z * se    # z = 1.96
```

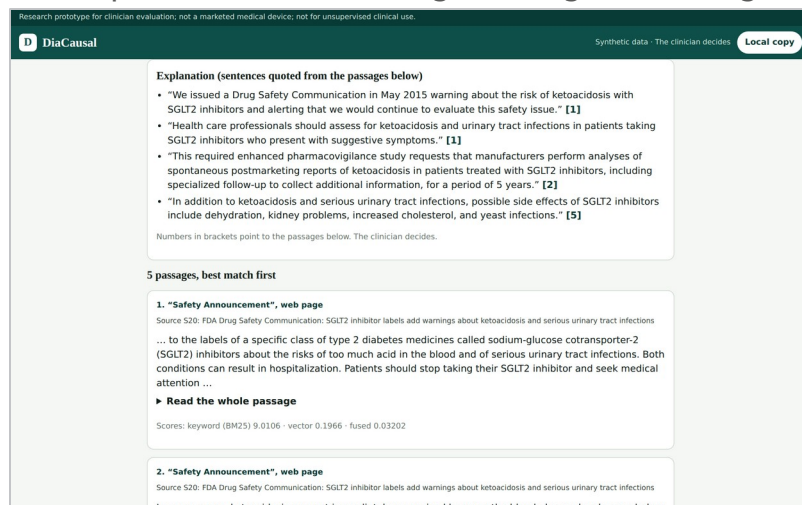
- **Safety thresholds live only in rules.csv**, each with a source
- **Output check** refuses any dose, or an estimate without its 95% range



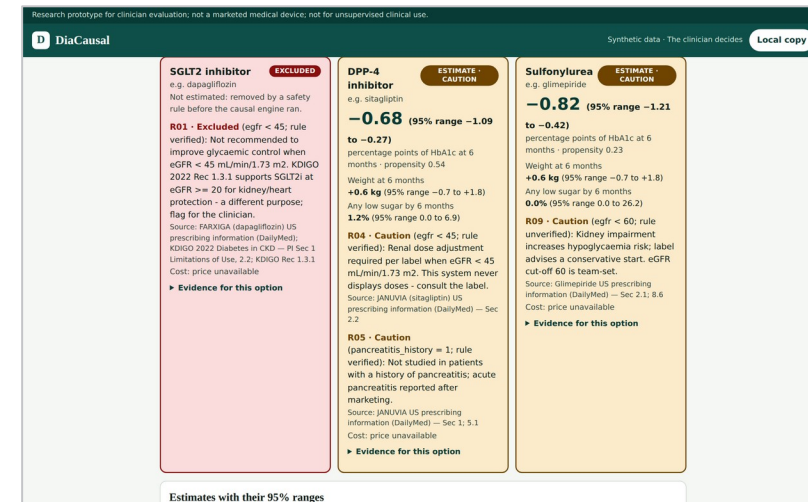
Implementation – Website Screens



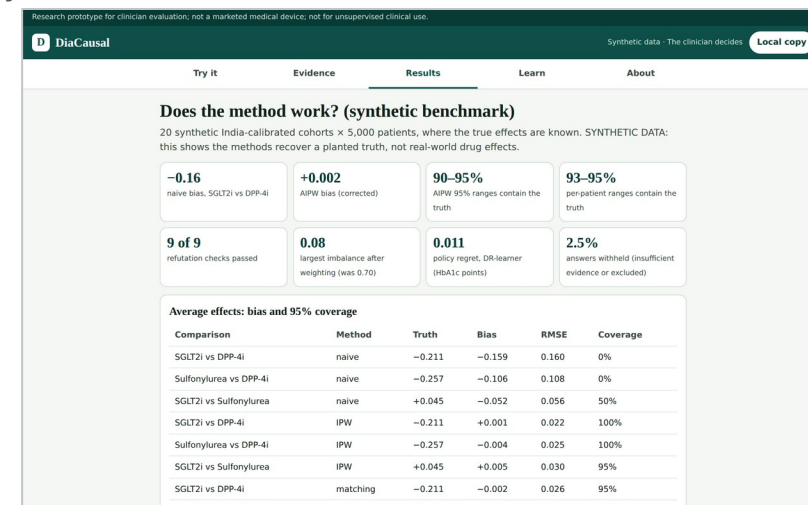
Try it: 3 options with 95% ranges, weight, low-sugar risk



Evidence: quoted sentences with citations [1][2]



Safety rules: excluded (red) and caution (amber), with sources



Results: the benchmark, on synthetic data



Implementation – Chat App Screens

DiaCausal [RESEARCH PROTOTYPE]

STEP 1 OF 2
Sign in
Second-line therapy support for adults already taking metformin.

User ID
dr.demo

Password

Sign in

[Trouble signing in?](#)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Type the 6 digits shown

25 9921

New image Play audio

Digits only. The audio reads the same 6 digits.

6 digits

Sign in: password + CAPTCHA, then an authenticator code

DiaCausal [RESEARCH PROTOTYPE]

Patient details [Example data]

New patient

Clears these details and the conversation.

Sex
Female Male

Age
18 or older.
58 years

Diabetes duration
6 years

HbA1c
8.4 %

eGFR
62 mL/min/1.73m²

BMI
31.2 kg/m²

Change in HbA1c at 6 months (percentage points; negative = better control)

SGLT2 inhibitor ESTIMATE	DPP-4 inhibitor ESTIMATE	Sulfonylurea ESTIMATE
-0.81 (95% range -0.91 to -0.71) Weight -2.4 kg (95% range -3.7 to -1.1 kg) Low-sugar risk 2.4% (95% range 0.6-5.3%) Similar patients who got it: propensity 0.55	-0.75 (95% range -0.87 to -0.63) Weight -0.3 kg (95% range -0.7 to +0.1 kg) Low-sugar risk 1.8% (95% range 0.6-5.7%) Similar patients who got it: propensity 0.33	-0.95 (95% range -1.09 to -0.81) Weight +1.3 kg (95% range +0.7 to +1.8 kg) Low-sugar risk 16.6% (95% range 8.6-25.2%) Similar patients who got it: propensity 0.11

SGLT2i minus DPP4i: **-0.06** (95% range -0.22 to +0.10) — the range includes 0, so neither is clearly better.

SU minus DPP4i: **-0.20** (95% range -0.39 to -0.01)

SGLT2i minus SU: **+0.14** (95% range -0.03 to +0.31) — the range includes 0, so neither is clearly better.

Estimates come from an India-calibrated synthetic cohort, not real patients. Decision support only. The clinician decides. Method: DR-learner on cross-fitted ADW scores, HC3 95% interval.

Evidence (sentences quoted from the passages below)

- “The objectives of these guidelines are: To consider the use of DPP-4 inhibitors, SGLT-2 inhibitors, and TZDs as second- and third-line treatment after metformin and sulfonylureas for controlling hyperglycaemia in type 2 diabetes in non-pregnant adults, including whether these oral agents are preferable to insulin.” [8]
- “Consider DPP-4 inhibitors as a possible cause for any patient who presents with severe and persistent joint pain, and consider discontinuation of therapy with this class of drugs.” [1]

Ask about a class, a dose, or a contraindication

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Estimates for this patient, each with a 95% range

DiaCausal [RESEARCH PROTOTYPE]

Dr Demo Sign out

Patient details [Example data]

New patient

Clears these details and the conversation.

Age
18 or older.
58 years

Diabetes duration
6 years

HbA1c
8.4 %

eGFR
62 mL/min/1.73m²

BMI
31.2 kg/m²

Category: Obese (≥25)
Asian-Indian cut-offs: normal <23 · overweight 23-24.9 · obese ≥25

Sulfonylurea: Safe to consider

Dummy reply from the DiaCausal backend. Your message (61 characters) arrived safely with trace ID cf8d6a43. No clinical analysis ran: the clinical guardrails, causal engine, guideline retrieval and explanation stages are not built yet.

3 of 6 stages ran in 54.2 ms; the others are not built yet.
Backend guard: passed · 2.2 ms Clinical guardrails: passed · 47.4 ms Causal engine: skipped
Guideline retrieval: skipped Explanation: skipped Output guard: passed · 4.6 ms

Trace ID cf8d6a43 — the same ID is in the browser console and the backend log.

YOU ASKED
Her mobile is [removed], which drug should she take?

Identifier removed — not sent
Please remove patient identifiers (Aadhaar, phone, PAN, email) and ask again.
Describe the patient by age, sex, lab values and current therapy instead.

Ask about a class, a dose, or a contraindication

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Guards: phone number removed, never sent

DiaCausal [RESEARCH PROTOTYPE]

Patient details [Example data]

New patient

Clears these details and the conversation.

Sex
Female Male

Age
18 or older.
58 years

Diabetes duration
6 years

HbA1c
8.4 %

eGFR
62 mL/min/1.73m²

BMI
31.2 kg/m²

Evidence (sentences quoted from the passages below)

- “Food and Drug Administration (FDA) safety review has found that type 2 diabetes medicines containing saxagliptin and alogliptin may increase the risk of heart failure, particularly in patients who already have heart or kidney disease.” [4]
- “Consider the risk and benefits of saxagliptin or alogliptin prior to initiating treatment in patients at a higher risk for heart failure.” [1]
- “Before taking saxagliptin or alogliptin, tell your health care professional if you have a history of heart failure or kidney impairment, as an increased risk of hospitalization for heart failure was found in clinical trials that included patients with these medical issues.” [2]
- “As a result, we have added new Warnings and Precautions to the labels of medicines that contain saxagliptin or alogliptin to inform of the potential increased risk of heart failure.” [3]

► The 5 passages and their sources

Estimated change in HbA1c at 6 months for this patient (percentage points; negative = better):

- SGLT2 inhibitor: -0.81 (95% range -0.91 to -0.71); weight -2.4 kg, low-sugar risk 2.4%.
- DPP-4 inhibitor: -0.75 (95% range -0.87 to -0.62); weight -0.3 kg, low-sugar risk 1.8%.
- Sulfonylurea: -0.95 (95% range -1.09 to -0.81); weight +1.3 kg, low-sugar risk 16.6%.

SGLT2i minus DPP4i: -0.06 (95% range -0.22 to +0.10); the range includes 0, so neither is clearly better.

SU minus DPP4i: -0.20 (95% range -0.39 to -0.01).

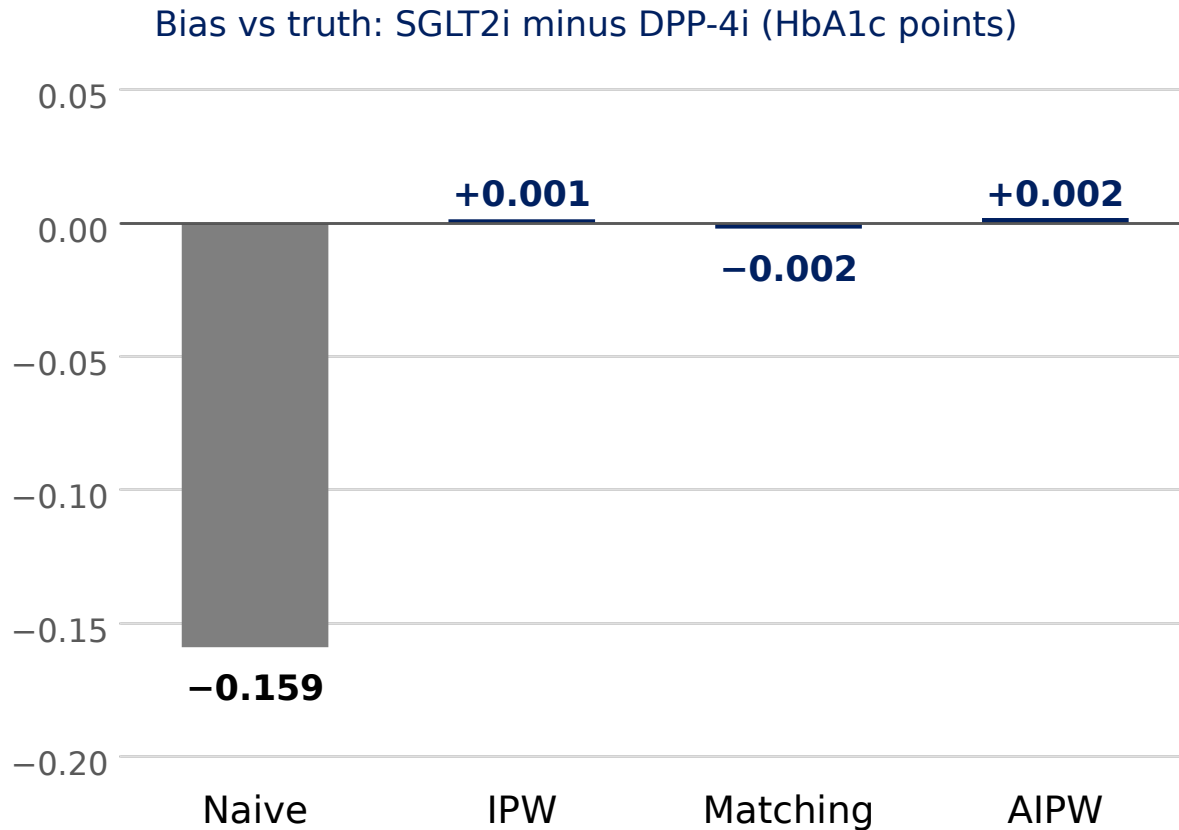
Ask about a class, a dose, or a contraindication

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Cited evidence: quoted sentences with passage numbers



Initial Results – Accuracy



-0.159 → +0.002

Bias: naive → AIPW

90-95%

AIPW 95% interval coverage

92.7-94.7%

DR-learner per-patient coverage

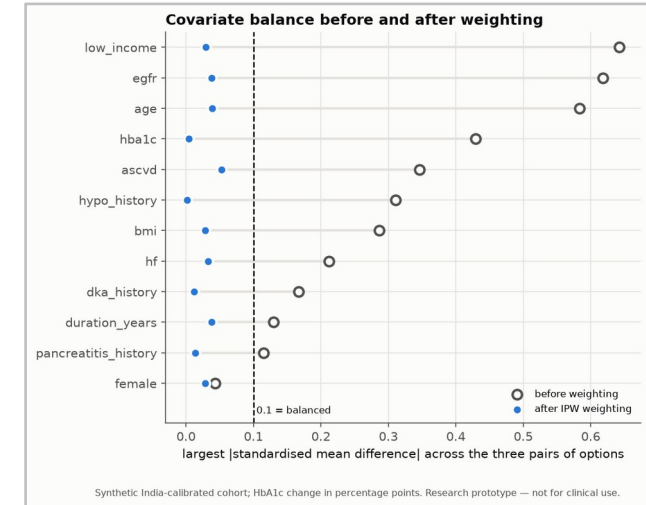
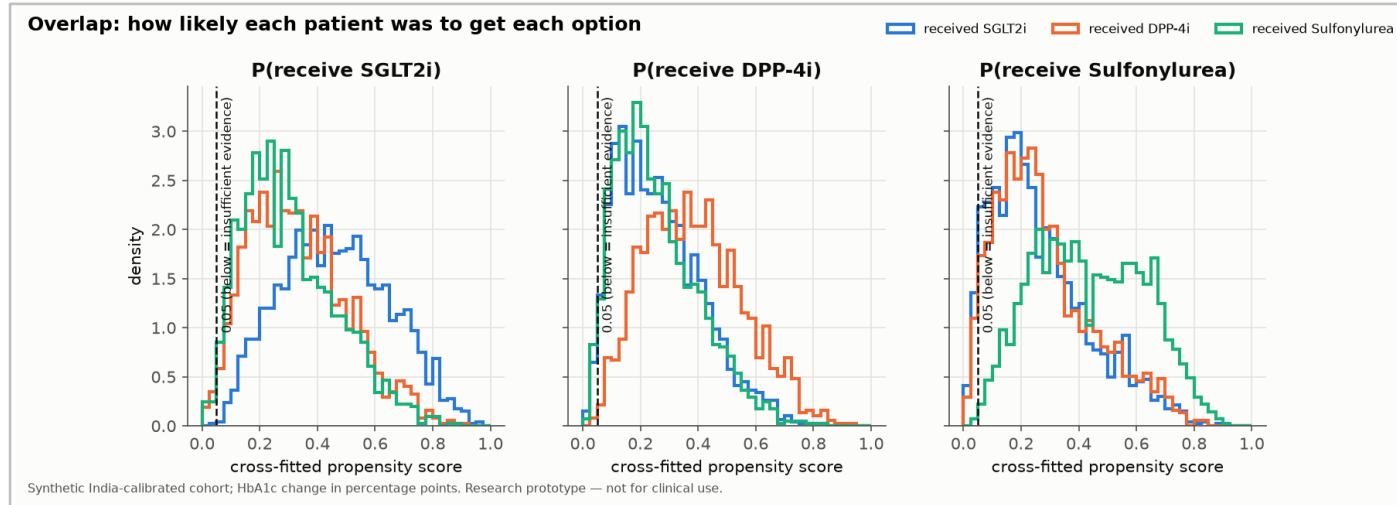
0.011

Policy regret (naive: 0.062)

20 synthetic India-calibrated cohorts × 5,000 patients; true effect -0.211 . Balance: largest SMD **0.70 → 0.08**. Refutation: **9 of 9** checks passed.

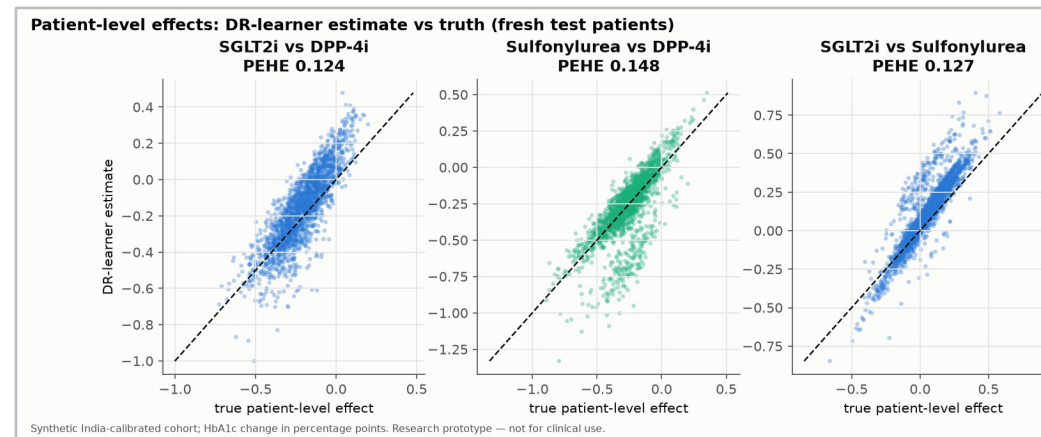
Synthetic data: shows the methods recover a planted truth, not real-world drug effects.

Initial Results – Plots



Overlap: below 0.05, the system abstains (“insufficient evidence”)

Balance: largest SMD 0.70 → 0.08



Per-patient effects vs truth on fresh test patients



Initial Results – Evidence Search (RAG)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

DiaCausal Synthetic data · The clinician decides **Local copy**

Explanation (sentences quoted from the passages below)

- “We issued a Drug Safety Communication in May 2015 warning about the risk of ketoacidosis with SGLT2 inhibitors and alerting that we would continue to evaluate this safety issue.” **[1]**
- “Health care professionals should assess for ketoacidosis and urinary tract infections in patients taking SGLT2 inhibitors who present with suggestive symptoms.” **[1]**
- “This required enhanced pharmacovigilance study requests that manufacturers perform analyses of spontaneous postmarketing reports of ketoacidosis in patients treated with SGLT2 inhibitors, including specialized follow-up to collect additional information, for a period of 5 years.” **[2]**
- “In addition to ketoacidosis and serious urinary tract infections, possible side effects of SGLT2 inhibitors include dehydration, kidney problems, increased cholesterol, and yeast infections.” **[5]**

Numbers in brackets point to the passages below. The clinician decides.

5 passages, best match first

1. “Safety Announcement”, web page
Source S20: FDA Drug Safety Communication: SGLT2 inhibitor labels add warnings about ketoacidosis and serious urinary tract infections ... to the labels of a specific class of type 2 diabetes medicines called sodium-glucose cotransporter-2 (SGLT2) inhibitors about the risks of too much acid in the blood and of serious urinary tract infections. Both conditions can result in hospitalization. Patients should stop taking their SGLT2 inhibitor and seek medical attention ...
► **Read the whole passage**
Scores: keyword (BM25) 9.0106 · vector 0.1966 · fused 0.03202

2. “Safety Announcement”, web page
Source S20: FDA Drug Safety Communication: SGLT2 inhibitor labels add warnings about ketoacidosis and serious urinary tract infections
In many cases, ketoacidosis was not immediately recognized because the blood glucose levels were below

0.933

Recall@5: right source in the top 5

1.000 · 0 leaks

Citation precision · dose leaks

0.800

Abstention when not covered (target 0.9)

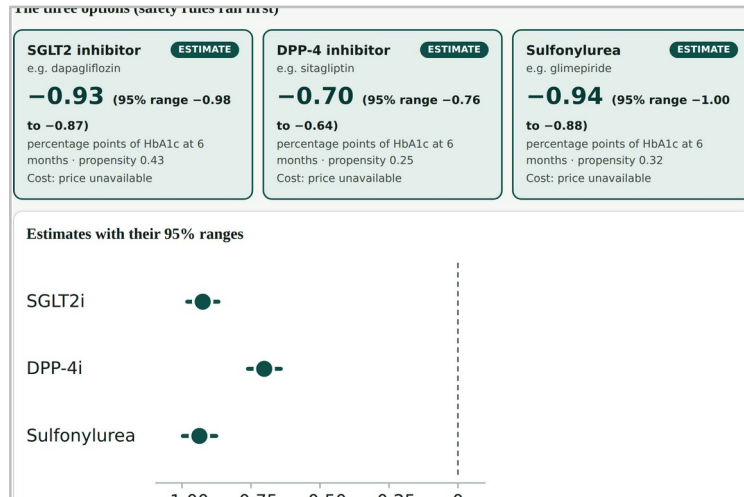
0.978

Answerable questions answered

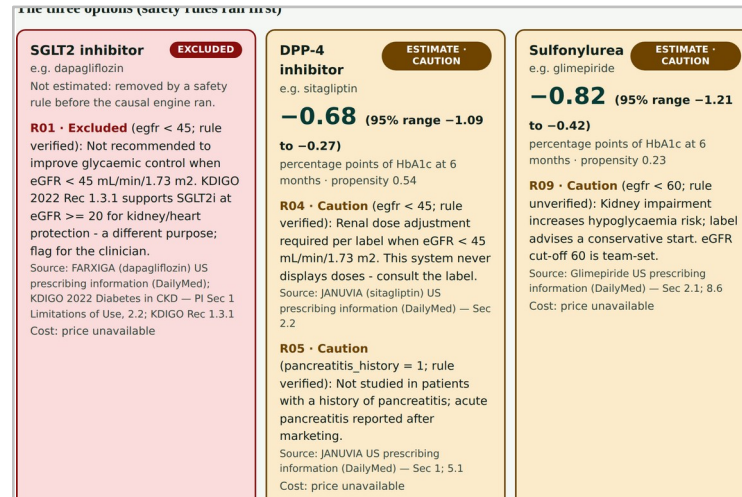
60 test questions (45 answerable, 10 out of scope, 5 dose) · results/rag_eval_summary.csv · 20 questions await a doctor’s review



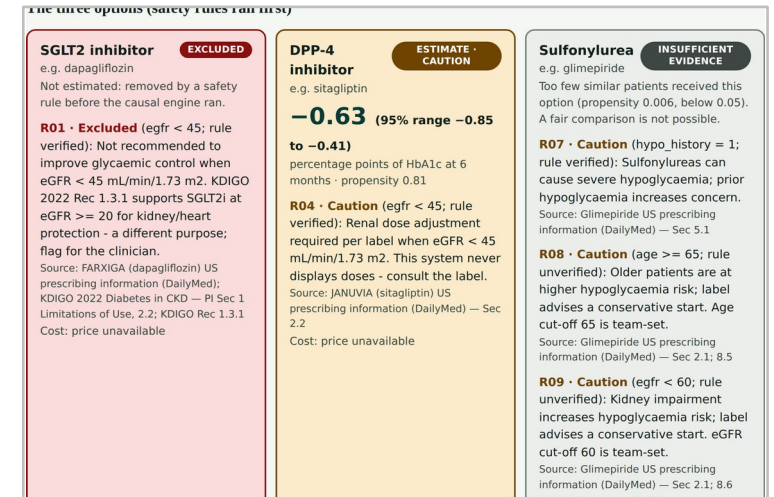
Demo



1 · Typical patient



2 · eGFR 40, past pancreatitis



3 · Older, past hypoglycaemia

Colour key (never green for “recommended”)

- Red** = excluded by a cited rule
- Amber** = caution
- Grey** = insufficient evidence
- Teal** = estimate with its 95% range

Live demo

Website (diacausal.netlify.app) or Streamlit, three preset patients

Backup

recorded demo video (112 s)



Work Completed vs Planned

Completed (by 30 Sept)

Problem, scope, literature and dataset research

Synthetic cohort, checked against real Indian data (NMB-2017)

Safety rules R01-R10; IPW, matching, AIPW, DR-learner

Weight and low-sugar risk; benchmark, 9 of 9 refutation checks

RAG: WHO + FDA sources, cited answers, 60-question test

Chat app runs all 6 stages: engine + evidence inside

Planned (interaction sheet)

Wk 8 · 7 Oct · 40%: met 28 Sep (Section 2)

Wk 9 · 14 Oct · 60%: met 29 Sep (Section 3)

Wk 10 · 21 Oct · 80%: better search (our target 3 Oct)

26-31 Oct: guide evaluation

Nov: final synopsis · Dec: 100% sign-off, research paper

Jan-Feb 2027: evaluation, PCUBE, paper, black book



Progress – About 75% of the Whole Project

Part	Share	Done	Evidence
Chat app: UI, guards, sign-in, patient panel, guardrails, voice	20%	~95%	401 backend + 246 frontend tests
Causal engine and benchmark	25%	~90%	150 tests; 9 of 9 checks. Missing: real data
RAG evidence search and explanations	20%	~75%	33 tests. Missing: embeddings, reranker
Joining engine + RAG + LLM in the chat app	15%	~85%	All 6 stages run (29 Sep)
Clinician evaluation (25 cases, SUS)	10%	~20%	Pack ready; not yet run
Report and IEEE paper	10%	~40%	Mid-sem report done
Total	100%	≈ 75%	$0.2 \times 95 + 0.25 \times 90 + 0.2 \times 75 + 0.15 \times 85 + 0.1 \times 20 + 0.1 \times 40$

Interaction sheet: 25% by 23 Sep (met), 40% by 7 Oct (met 28 Sep), 60% by 14 Oct (met 29 Sep). Shares are the team's estimate of each part of the year; each "done" points to tests or files.

Conclusion



- 1 **Causal engine and cited evidence, in one chat app**
- 2 **Safety rules run before** any estimate
- 3 **Honest uncertainty:** 95% ranges and refusal to guess
- 4 **Next:** better search model, doctor review, IEEE paper

The doctor always decides. The system supports, and never replaces, clinical judgement.



References

- [1] B. M. Shields et al., “Patient stratification for determining optimal second-line and third-line therapy for type 2 diabetes: the TriMaster study,” *Nat. Med.*, vol. 29, pp. 376–383, 2023.
- [2] J. M. Dennis et al., “Development of a treatment selection algorithm for SGLT2 and DPP-4 inhibitor therapies in people with type 2 diabetes,” *Lancet Digit. Health*, vol. 4, no. 12, pp. e873–e883, 2022.
- [3] L. M. Gudemann et al., “Validation of an algorithm for selection of SGLT2 and DPP4 inhibitor therapies in people with type 2 diabetes across major UK ethnicity groups,” *Lancet Reg. Health Eur.*, vol. 61, 101547, 2026.
- [4] R. M. Anjana et al., “Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17),” *Lancet Diabetes Endocrinol.*, vol. 11, no. 7, pp. 474–489, 2023.
- [5] E. H. Kennedy, “Towards optimal doubly robust estimation of heterogeneous causal effects,” *Electron. J. Stat.*, vol. 17, no. 2, pp. 3008–3049, 2023.
- [6] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. NeurIPS*, 2020.
- [7] A. G. Cashin et al., “Transparent reporting of observational studies emulating a target trial – the TARGET statement,” *JAMA*, vol. 334, no. 12, pp. 1084–1093, 2025.
- [8] R. Nagarathna et al., “A composite of BMI and waist circumference may be a better obesity metric in Indians with high risk for type 2 diabetes (NMB-2017),” *Diabetes Res. Clin. Pract.*, 2020.
- [9] World Health Organization, “Guidelines on second- and third-line medicines and type of insulin for the control of blood glucose levels in non-pregnant adults with diabetes mellitus,” WHO, Geneva, 2018.



Thank You!



Backup – Why Synthetic Data Is Valid

2. The same patients with their hidden answers (possible only because the data is synthetic)

age	hba1c	egfr	treatment	y	y_true_SGLT2i	y_true_DPP4i	y_true_SU	e_true_SGLT2i	e_true_DPP4i	e_true_SU
56	7.70	52	SU	-1.38	-1.23	-1.27	-1.38	0.12	0.66	0.21
54	9.70	85	SGLT2i	-1.55	-1.55	-1.24	-1.53	0.32	0.15	0.53
61	9.80	76	SU	-1.07	-0.81	-0.54	-1.07	0.49	0.25	0.26
56	9.70	85	SGLT2i	-0.37	-0.37	-0.06	-0.51	0.42	0.24	0.34
50	9.70	112	SGLT2i	-1.76	-1.76	-1.28	-1.67	0.88	0.04	0.08
59	8.90	102	DPP4i	-0.90	-1.25	-0.90	-1.28	0.61	0.16	0.22
68	9.10	78	SGLT2i	-1.12	-1.12	-0.90	-1.30	0.41	0.37	0.21
64	9.30	66	SU	-1.87	-1.72	-1.55	-1.87	0.43	0.38	0.19

Shaded columns are never shown to the estimators. y_true_X = the 6-month HbA1c change this patient WOULD have had on drug X (y equals the one for the drug actually given); e_true_X = the true chance a doctor would pick X for this patient. Knowing these is what lets the benchmark grade each method.

- Real records never show what would have happened on the other drug, so no real dataset can grade a causal method.
- Synthetic data with a known truth is how causal methods are tested (ACIC challenges, IHDP benchmark).
- Ours: truth -0.211 ; naive -0.370 ; AIPW -0.209 with 95% coverage; 9 of 9 refutation checks.
- It proves the method, not the drug effects: 117 numbers = 9 cited, 15 assumed-directional, 93 team-set.



Backup – Real Indian Diabetes Datasets

Dataset	What it has	Drug + later HbA1c?	Access
NMB-2017 (Nagarathna 2020)	7,496 adults; HbA1c, BMI, waist, family history	No	Open, CC0 (in our repo)
LASI Wave 1 (IIPS)	~66,000 adults 45+; HbA1c from dried blood spot	No	Free request with ID
NFHS-5 (2019-21)	Blood glucose; on diabetes medicine or not	No	Registration (DHS Program)
CARRS (Chennai, Delhi)	12,271 adults, followed every 2 years	No add-on drug	Request to investigators
ICMR-INDIAB	National prevalence (~113,000)	No	Published tables only
Pune clinic study (2025)	18,950 T2D patients; HbA1c, medicines	Partly	On request to authors

Pima is Native American (US) data, not Indian, and has no drug or outcome: not used. Checked 28 Sept 2026.



Backup – From Prototype to a Real Hospital

- 1 Data: hospital records of metformin patients who started SGLT2i, DPP-4i or a sulfonylurea, with HbA1c 3–9 months later
- 2 Approvals: hospital ethics committee, de-identification, India's DPDP Act 2023
- 3 Same code: the records have the same columns as our synthetic cohort, so nothing changes
- 4 Checks without a known truth: overlap rule, refutation tests, E-values, agreement with trial results
- 5 Silent pilot: doctors see the suggestion but decide as usual; later a CDSCO medical-device review

“We proved the method where the truth is known; next, the same code on ethically approved hospital records.”

Backup – Phone Screens



Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

D DiaCausal Local copy

SGLT2 inhibitor ESTIMATE
e.g. dapagliflozin

−0.93 (95% range −0.98 to −0.87)
percentage points of HbA1c at 6 months · propensity 0.43
Weight at 6 months
−2.1 kg (95% range −2.3 to −2.0)
Any low sugar by 6 months
2.5% (95% range 0.8 to 4.2)
Cost: price unavailable
► Evidence for this option

DPP-4 inhibitor ESTIMATE
e.g. sitagliptin

−0.70 (95% range −0.76 to −0.64)
percentage points of HbA1c at 6 months · propensity 0.25
Weight at 6 months
−0.4 kg (95% range −0.6 to −0.2)
Any low sugar by 6 months
1.8% (95% range 0.1 to 3.4)
Cost: price unavailable

Try it Evidence Results Learn About

Try it

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

D DiaCausal Local copy

Explanation (sentences quoted from the passages below)

- “We issued a Drug Safety Communication in May 2015 warning about the risk of ketoacidosis with SGLT2 inhibitors and alerting that we would continue to evaluate this safety issue.” [1]
- “Health care professionals should assess for ketoacidosis and urinary tract infections in patients taking SGLT2 inhibitors who present with suggestive symptoms.” [1]
- “This required enhanced pharmacovigilance study requests that manufacturers perform analyses of spontaneous postmarketing reports of ketoacidosis in patients treated with SGLT2 inhibitors, including specialized follow-up to collect additional information, for a period of 5 years.” [2]

Try it Evidence Results Learn About

Evidence

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

D DiaCausal Local copy

Does the method work? (synthetic benchmark)

20 synthetic India-calibrated cohorts × 5,000 patients, where the true effects are known. SYNTHETIC DATA: this shows the methods recover a planted truth, not real-world drug effects.

−0.16 naïve bias, SGLT2i vs DPP-4i	+0.002 AIPW bias (corrected)
90–95% AIPW 95% ranges contain the truth	93–95% per-patient ranges contain the truth
9 of 9 refutation checks passed	0.08 largest imbalance after weighting (was 0.70)
0.011 policy regret, DR-	2.5% answers withheld

Try it Evidence Results Learn About

Results

DiaCausal PROTOTYPE Sign out

Patient details Example data

New patient

Clears these details and the conversation.

Age
18 or older.
58 years

Diabetes duration
6 years

HbA1c
8.4 %

eGFR
62 mL/min/1.73m²

BMI
31.2 kg/m²

Category: Obese (≥25)

Asian Indian, cut-off: normal <22, overweight 22–24.9

Chat: patient panel

DiaCausal PROTOTYPE Sign out

Patient details Example data

Dummy reply from the DiaCausal backend. Your message (61 characters) arrived safely with trace ID 726a82bf. No clinical analysis ran: the clinical guardrails, causal engine, guideline retrieval and explanation stages are not built yet.

3 of 6 stages ran in 33.0 ms; the others are not built yet.
Backend guard: **passed** · 1.4 ms
Clinical guardrails: **passed** · 29.1 ms
Causal engine: skipped
Guideline retrieval: skipped
Explanation: skipped
Output guard: **passed** · 2.5 ms

Trace ID 726a82bf — the same ID is in the browser console and the backend log.

Ask a question

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

Chat: reply



Backup – Literature Comparison

Paper	Data	Approach	Gap for DiaCausal
Shields et al. 2023 (TriMaster)	RCT, UK	Three-way crossover trial	Two simple strata; UK population
Dennis et al. 2022	UK primary care	SGLT2i vs DPP-4i selection model	Two drugs; no sulfonylurea; not Indian
Güdemann et al. 2026	UK primary care	Validation across ethnic groups	Smaller South Asian benefit; recalibrate
Quaranta and Pietrantuono 2025	Pima (768)	Causal discovery, DoWhy	Not Indian; no treatment variable
Hasan et al. 2025	MIMIC-III	Causal trees	Single ICU dataset; not validated
Zakka et al. 2024 (Almanac)	314 clinical questions	RAG over curated sources	No patient-specific causal estimates
Xiong et al. 2024 (MIRAGE)	Benchmark	Benchmarking medical RAG	Question answering, not decision support

Week 1 of the interaction sheet: 16 papers reviewed; the most relevant are compared here and in the report (Chapter 2).